

Key responsibilities of a PCB designer

1. Schematic entry
2. Footprint creation
3. Proper component placement
4. Track-width selection
5. Routing
6. Creating NC/Gerber data
7. Ensuring DRG, design for manufacturability (DFM), design for assembly (DFA)
8. Knowing IPC/Mil standards
9. Knowing of electromagnetics, electromagnetic compatibility (EMI)/electromagnetic interference (EMC) fundamentals

You just need to find out where they place the buttons." I believe that if you are a good designer on one CAD system, with a slight learning curve, you should be able to pick up another system with no problem," Derek told PCBDesign007 in an interview.

Who's hiring?

It is difficult to point out a particular industry as the major recruiter of PCB designers. Aspirants in this field may look for opportunities in all industries which are directly involved in manufacturing electronics products. These include telecom, automotive, biomedical equipment, computer hardware, electronic security systems, consumer electronic appliances, mobile compa-

nies, industrial automation and power electronics, amongst others.

"The big organisations always have a team of six-eight designers while the middle ones have two-five designers. The team size depends upon the development of new hardware or modification of the existing one, and also on the size of the R&D team. However, to name a few, some of the major recruiters in this field are Bharat Electronics Limited, Semiconductor Complex Laboratory, HCL, L&T, GE, HP, Beitel, Nokia, Samsung, LG, Ericsson, Sony and Philips," informs Singla.

Krithiga adds, "Small-sector PCB design companies are recruiting freshers. MNCs like Wipro, Applied Materials, TCS, Ingersoll Rand, and most of the electronics goods designing companies offer a good opportunity for PCB design engineers."

Getting the job signal

Recruiters look for electronic and information engineering (EIE), EEE and ECE engineers with basic electronics components knowledge with any board design software like OrCAD, Allegro, Altium or PADS. At present, most of the companies are using OrCAD and Allegro for circuit board design.

A PCB designer's 'must haves' as

stated by Singla are as follows:

1. Understanding of the functioning of an electronic circuit.
2. Knowledge of IPC/Mil standards.
3. Know the process of PCB manufacturing, assembling and testing.
4. Good knowledge of various electronic components' packaging.
5. Know about different PCB base materials.
6. Able to interact with circuit designers, PCB manufacturing and assembling team.
7. Knowledge of EMI/EMC.

If the PCBs are not designed properly, one runs the risk of facing a high level of rework/rejection of populated or unpopulated PCBs. "I have also seen a lot of circuit malfunctions/failures in the field, which are a result of poor design, fabrication or soldering techniques," Singla says. He also suggests that PCB designers should have a detailed knowledge of not only PCB fabrication but also soldering techniques. Also, they are required to be kept abreast of the limitations of PCB manufacturers with respect to minimum drill-size, minimum track-width, minimum track-to-track and track-to-pad distance, tolerances, etc. ●

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